

Gaussian Mixture Model using Expectation-Maximization (EM) Algorithm

Project Objective

The objective of this project is to implement the Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) Algorithm from scratch in Base R. The project demonstrates how GMM performs probabilistic clustering by assigning each observation a probability of belonging to each cluster. The implementation is completed without using any external machine learning or clustering libraries.

Software Used

- R
 - Base R
 - R GUI / RStudio
 - Git
 - GitHub
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Dataset

A synthetic dataset containing **20,000 observations** was generated using Base R.

Features:

- X1
- X2

The dataset contains two Gaussian clusters and is saved as:

synthetic_data.csv

Steps to Run the Program

1. Open **gmm_em_algorithm.R** in R GUI or RStudio.
2. Run the complete program.
3. The synthetic dataset is generated and saved as **synthetic_data.csv**.
4. The EM Algorithm initializes the model parameters.
5. The Gaussian Density Function is calculated.
6. The Expectation (E-Step) computes the responsibility matrix.
7. The Maximization (M-Step) updates the model parameters.
8. The algorithm repeats until convergence or 20 iterations.

9. The following output plots are generated inside the **Output** folder:

- dataset.png
- final_clusters.png
- log_likelihood.png

Project Folder Structure

GMM-EM-Algorithm-Week1/

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├— README.md

├— gmm_em_algorithm.R

├— synthetic_data.csv

├— Week1_Report.pdf

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└─ Output/

├— dataset.png

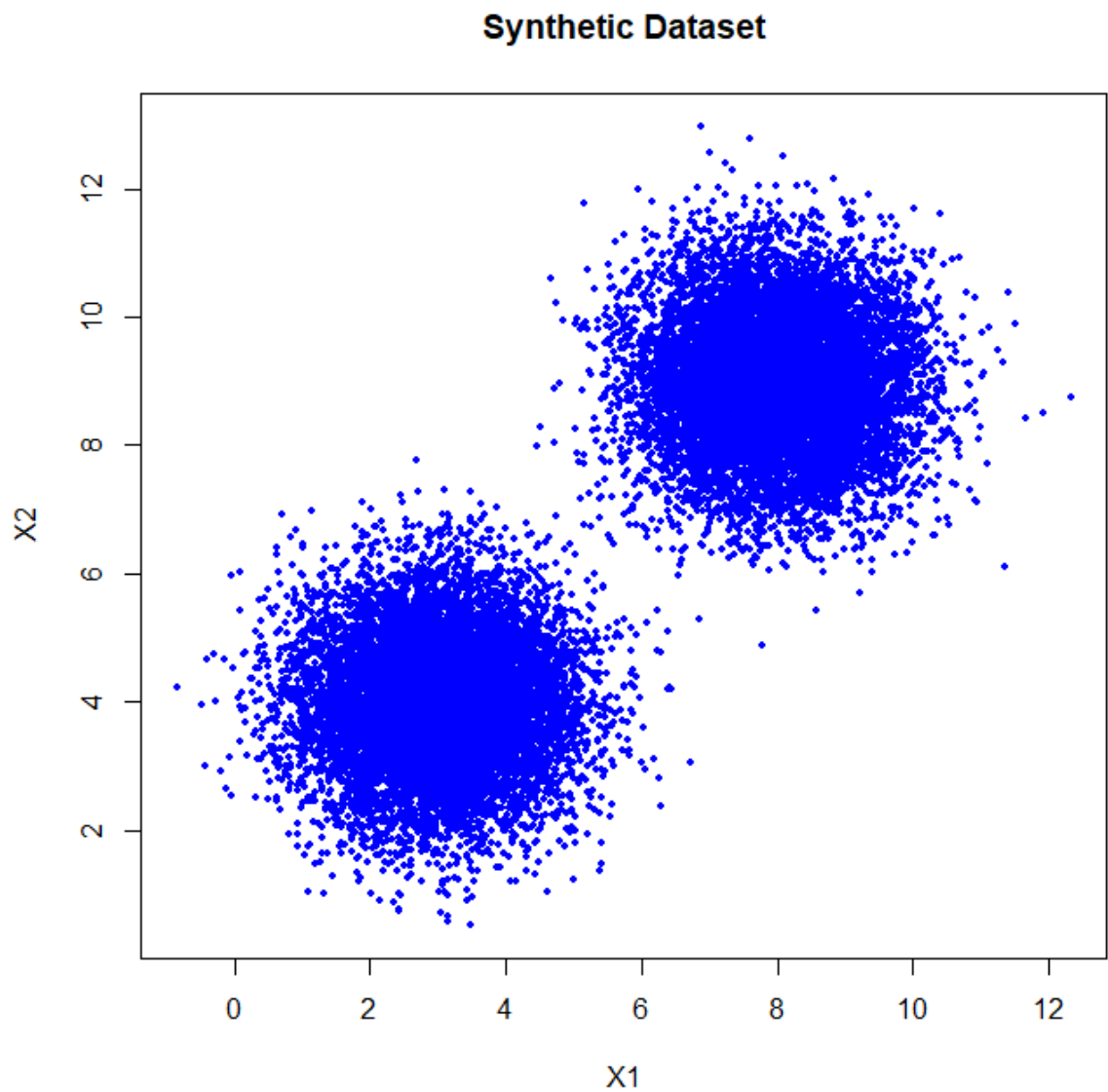
├— final_clusters.png

└─ log_likelihood.png

Output Screenshots

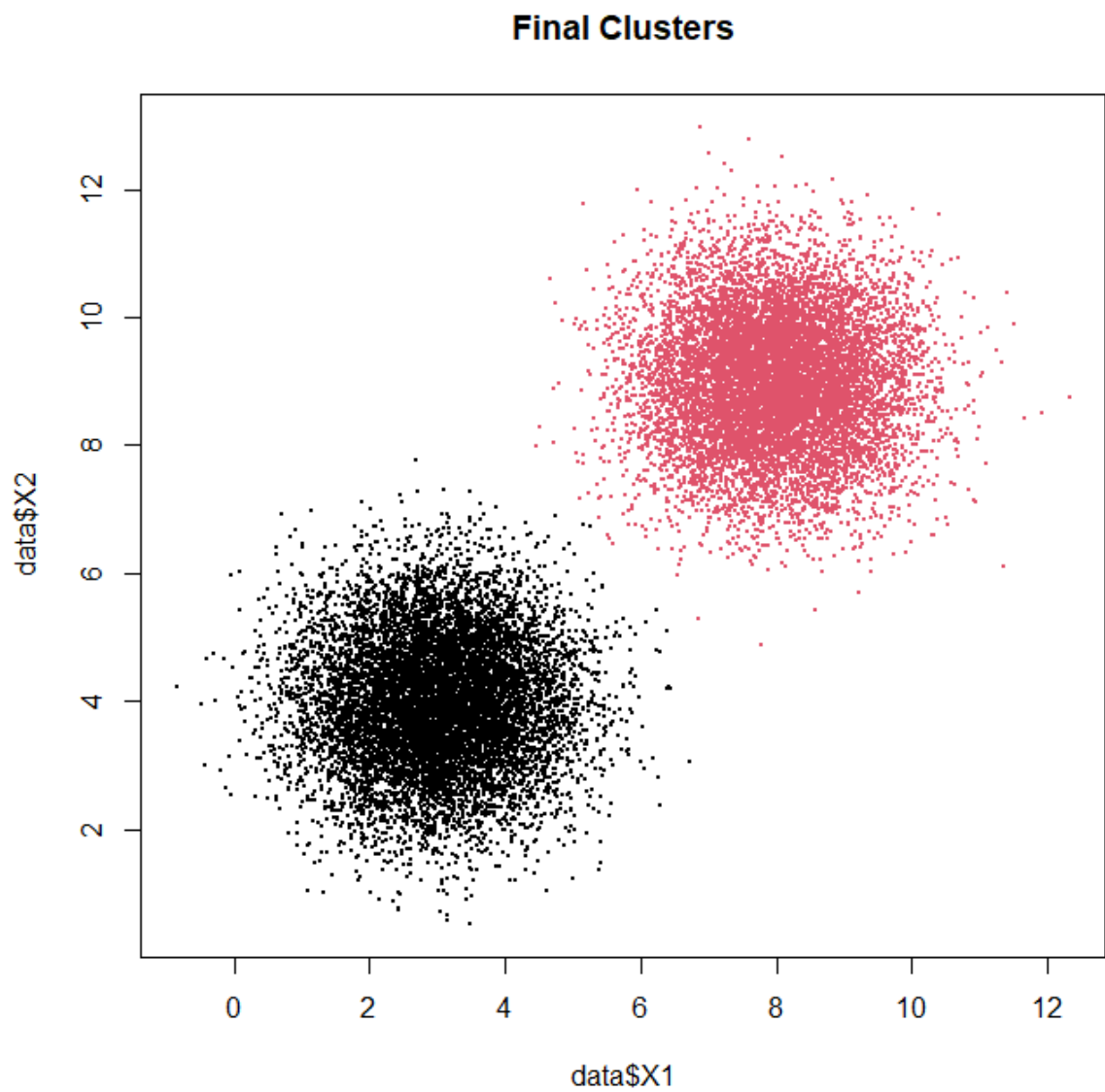
1. Synthetic Dataset

- Scatter plot of the generated Gaussian clusters.



2. Final Cluster Assignment

- Scatter plot showing the clusters identified by the EM Algorithm.

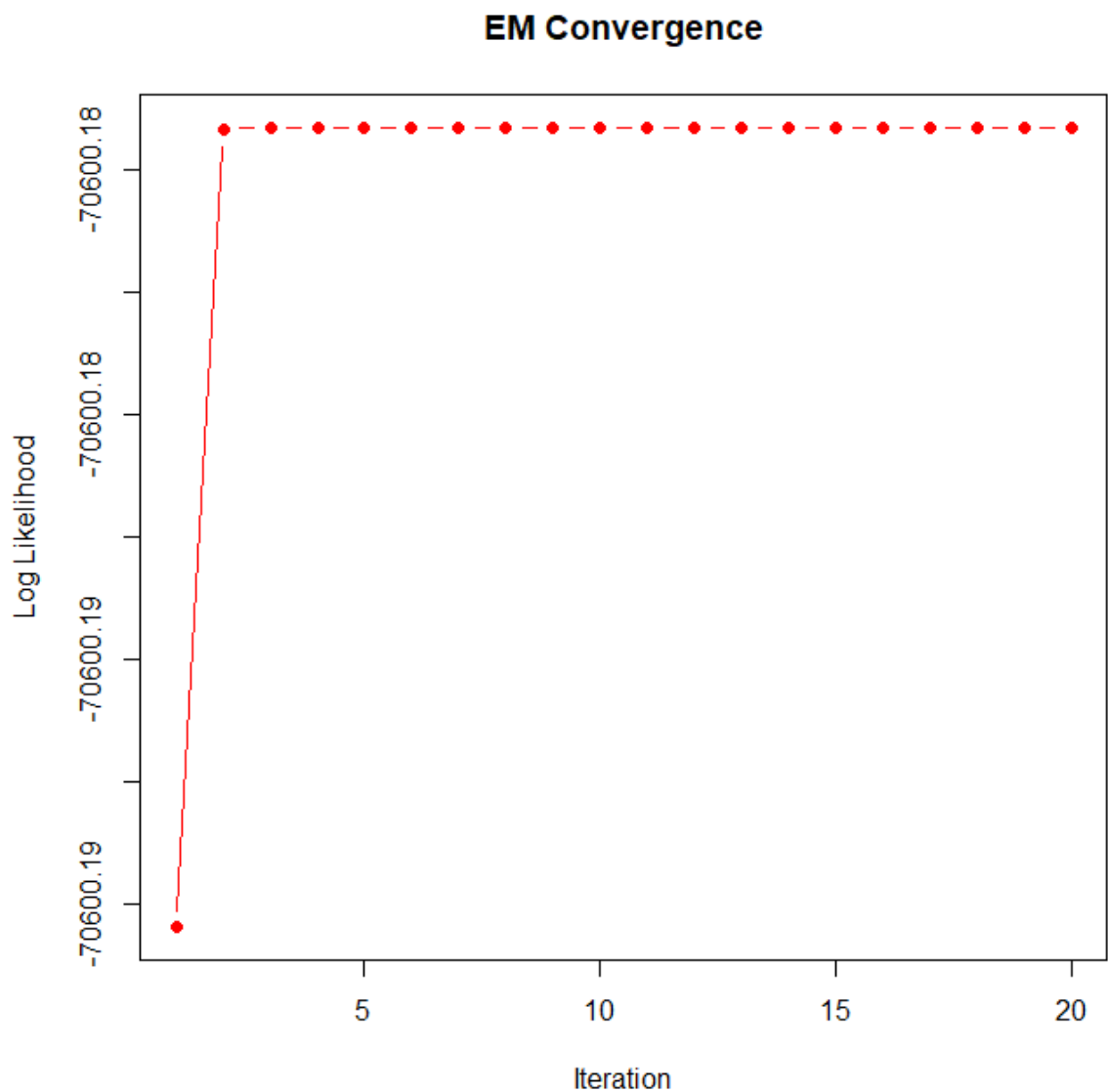


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3. Log-Likelihood Plot

- Graph showing the Log-Likelihood value for each iteration until convergence.

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Learning Outcomes

After completing this project, I learned to:

- Understand the concept of unsupervised learning.
- Differentiate between K-Means and Gaussian Mixture Models.
- Generate a synthetic dataset using Base R.
- Implement the Multivariate Gaussian Probability Density Function.
- Perform the Expectation (E-Step).
- Perform the Maximization (M-Step).
- Update mixing coefficients, mean vectors, and covariance matrices.
- Compute Log-Likelihood values.

- Check the convergence of the EM Algorithm.
 - Visualize clustering results using Base R.
 - Upload the completed project to GitHub.
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Author

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Course: BCA

Project: Week 1 – Gaussian Mixture Model using EM Algorithm

Language: R (Base R)